```
# data dividation
# workflow of ml
# data ingestion==>data cleaning and preprocessing -->based on the target co
# we have to select ml model==>model tranining ==>model prediction ==>model
# evaluation
```

```
# df===>x(input),y(target)
# x==>x_train,x_test
# y===>y_train,y_test
# model traning(x-train,y_train)
# y_pred=model prediction(x_test)
# model performence(y_test,y_pred)
```

```
# 1+1=2
# 1+2=3
# 1+3=4
# 1+4=5
# x=[1+1,1+2,1+3,1+4]
# y=[2,3,4,5]
# x_train=[1+1,1+2,1+3]
# y_train=[2,3,4]
# x_test=[1+4]
# y_test=[5]
```

```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('/content/placement.csv')
df.head()
```

| | cgpa | resume_score | placed |
|---|---|---|---|
| **0** | 8.14 | 6.52 | 1 |
| **1** | 6.17 | 5.17 | 0 |
| **2** | 8.27 | 8.86 | 1 |
| **3** | 6.88 | 7.27 | 1 |
| **4** | 7.52 | 7.30 | 1 |

Next steps: Generate code with df | New interactive sheet

```
df.shape
```

```
(100, 3)
```

```
x=df.drop(columns=['placed'])
#x
y=df['placed']
#y
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_stat
```

```
print('total shape of df:',df.shape)
print('------****-----')
print('total input data shape',x.shape)
print('x_train data shape',x_train.shape)
print('x_test data shape',x_test.shape)
print('----')
print('total output data shape',y.shape)
print('y_train data shape',y_train.shape)
print('y_test data shape',y_test.shape)
```

```
total shape of df: (100, 3)
------****-----
total input data shape (100, 2)
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1787/2072330234.py in <cell line: 0>()
      2 print('------****-----')
      3 print('total input data shape',x.shape)
----> 4 print('x_train data shape',x_train.shape)
      5 print('x_test data shape',x_test.shape)
      6 print('----')

NameError: name 'x_train' is not defined
```

Next steps: Explain error

Double-click (or enter) to edit

Close

Double-click (or enter) to edit

```
# df=(100,3)=x(100,2)+y(100,1)
# x(100,2)=x_train(80,2)+x_test(20,2)
# y(100,1)=y_train(80,1)+y_test(20,1)
```

```
# task=covid dataset convert into categorical to numeri
# df=x,y,x_train,x_test,y_train,y_test
```

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```